

PCI PCL-AI · COURSE OUTLINE

Finance and delivery. One profession.

The syllabus for the AI Project Controls Leader — the credential that examines the accounting **and** the arithmetic of delivery, because a project needs both and almost nobody is trained in both.

13 domains · 61 knowledge areas →

WHY THIS CREDENTIAL EXISTS

Two professions. One project.

THE ACCOUNTANT

Knows the money

- When revenue may be recognised
- What a provision must satisfy
- Which costs may be capitalised

Rarely examined on a critical path, an earning rule, or why a schedule slip becomes a cost.

THE ENGINEER

Knows the work

- Where the critical path runs
- What the float really is
- How progress is measured

Rarely examined on cut-off, a contract asset, or the difference between billed and earned.

A project lives in the overlap. **The overlap is where the money is lost**, and it is the one place neither training looks.

WHAT THE OVERLAP COSTS

A CPI of 1.19 and 1.05 from the same month

Earned value 2,200,000. Invoiced cost 1,850,000. Skip the accrual and **CPI = 1.19**. Book the 240,000 already done and **CPI = 1.05**.

The error is **accounting** — a cut-off failure. The damage is **delivery** — a performance index a board will act on. Fourteen points, one missing entry.

An accountant who never reads a CPI will not see it. A planner who never raises an accrual will not see it either. **That is the case for this credential in one month-end.**

THE SYLLABUS

Forty, forty, twenty

13

domains

61

knowledge areas

26

sector case studies

40% finance, accounting and reporting · **40%** project management · **20%** governed
AI. The proportions of the Body of Knowledge.

THE FINANCE HALF

Standards a controls lead is examined on

IFRS 15

revenue from contracts with customers — the five-step model, over-time recognition, variable consideration and the constraint

IAS 37

provisions and contingent liabilities — behind every warranty reserve, onerous-contract charge and dispute disclosure

IAS 2

the materials and work-in-progress a project holds in store

IAS 16 / IFRS 16

the plant a project owns, and the plant and premises it leases

IAS 23

the financing cost of building a major asset

IAS 1

presentation of financial statements

IAS 11

the construction-contracts standard IFRS 15 replaced — covered so legacy terminology in an older file still reads

DOMAIN 2 · REVENUE

The recognition chain, in full

Percentage of completion (PoC) = Costs incurred to date / Total estimated costs

over-time recognition by the input method

Cumulative revenue = PoC × Transaction price

revenue earned to date

Period revenue = Cumulative revenue – revenue recognised in prior periods

what lands in this month

Contract asset (liability) = cumulative revenue – cumulative billed

over- or under-billing, on the balance sheet

Recognition is not billing. A project can be profitable, fully billed and **still carry a contract liability**.

Before the number exists

Assets = Liabilities + Equity

the identity every ledger satisfies

catch-up = PoC × revised price – revenue recognised to date

the cumulative catch-up when the price is revised

Revised transaction price = fixed price + constrained variable consideration

variable consideration, constrained

Capitalised borrowing cost = weighted-average qualifying expenditure × capitalisation rate

IAS 23 — financing cost inside the asset

NRV = estimated selling price – costs to complete and sell

IAS 2 — the write-down test on stored materials

Contract to cash

Amount = quantity × rate

the bill-of-quantities line, before anything else happens

amount due = gross value – retention – previous payments

the payment application

Fee = target fee + contractor's share × (target cost – actual cost)

target-cost pain/gain share

LD exposure = LD rate × forecast days late

liquidated damages, forecast not incurred

CCC = DSO + DIO – DPO

cash conversion cycle, in days

DOMAIN 6 · EARNED VALUE

The delivery half

$CV = EV - AC$

cost variance

$SV = EV - PV$

schedule variance, in money

$CPI = EV / AC$

cost performance index

$SPI = EV / PV$

schedule performance index

$SV(t) = ES - AT$

earned schedule variance, in time

$SPI(t) = ES / AT$

the index that still works near completion

DOMAIN 6 · THE HEART OF IT

There is no ‘the’ EAC

$$\text{EAC} = \text{AC} + (\text{BAC} - \text{EV})$$

the overrun was one-off; the rest runs to plan

$$\text{EAC} = \text{BAC} / \text{CPI}$$

performance to date persists — usually the honest default

$$\text{EAC} = \text{AC} + (\text{BAC} - \text{EV}) / (\text{CPI} \times \text{SPI})$$

cost and schedule pressure both continue

$$\text{EAC} = \text{AC} + \text{ETC}$$

the team re-estimated the remaining work bottom-up

Four methods. Four assumptions. Four different answers from the same data. **Naming which one you used, and why, is the skill.**

DOMAINS 4, 9, 12

Why the number moved

Price/rate = (actual price – standard price) × actual quantity

price variance

usage/efficiency = (actual quantity – standard quantity) × standard price

usage variance — the other half of the story

EV = (points done / total points) × BAC

earned value when scope is a backlog

EMV = probability × impact

expected monetary value — how risk enters a forecast

PART ONE · 40%

Finance, accounting and reporting

1

5 KAs

Foundations of accounting for project controls

The accounting model · Components of the financial statements · Accrual accounting and the matching concept · Cost provisions and cost accruals · Chart of accounts and cost coding for projects

2

5 KAs

Financial reporting and the standards

The reporting framework · IFRS 15 Revenue from Contracts with Customers · Revenue recognition beyond the basics · Other relevant standards for project controls · Management reporting versus statutory reporting

3

5 KAs

Budgeting and forecasting

Budgeting fundamentals · Cost estimation · The time-phased budget, or cost baseline (planned value) · Forecasting · Cash-flow forecasting

4

4 KAs

Performance management, variance analysis and management reporting

Performance management principles · Variance analysis · Management reporting · Data visualisation and storytelling for controls

PART TWO · 40%

Project management

5	Cost management and cost control
4 KAs	The cost management framework · The cost control cycle · Cost breakdown and control accounts · Change control and cost impact
6	Earned value management and forecasting
4 KAs	EVM fundamentals · Variances and performance indices · Forecasting with EVM: the EAC family · Integrating cost and schedule, limitations, earned schedule
7	Contracts, commercial management, BoQ, invoicing and revenue
5 KAs	Types of contract · Contract management · Bills of quantities · Invoicing and applications for payment · Revenue recognition in the commercial cycle
8	Project management lifecycle
6 KAs	Initiating · Planning · Executing · Monitoring and controlling · Closing · Development approaches: predictive, iterative, incremental and adaptive

PART TWO · CONTINUED

Delivery, schedule, process and risk

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| 9
6 KAs | Agile, Scrum and adaptive delivery for project controls
Agile foundations · The Scrum framework in depth · Backlogs, estimation and agile metrics · Kanban, Lean and scaling · Agile cost control, forecasting and earned value · Hybrid delivery and agile governance |
| 10
4 KAs | Project scheduling, in depth
Schedule development · Network analysis and the critical path method · Schedule compression and resourcing · Progress measurement and schedule control |
| 11
3 KAs | Business process cycles
Order-to-cash · Procure-to-pay · Internal control and segregation of duties |
| 12
3 KAs | Risk management for project controls
The risk framework · The risk process · Contingency and management reserve |
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PART THREE · 20%

AI, governed

13 AI for project controls and project management

7 KAs

AI foundations for professionals · Data: the fuel · Prompting and working with generative AI · AI tool categories for project controls and PM · AI applied across the project-controls lifecycle · Governance, ethics, risk and assurance of AI · Building an AI-augmented project-controls capability

$$F1 = 2 \times (\text{precision} \times \text{recall}) \div (\text{precision} + \text{recall})$$

whether the model is worth running at all

A model with 99% recall and 4% precision buries the team in false positives. **“The AI found something” is not a control.**

BEHIND THE SYLLABUS

113 standards. 532 process requirements.

Each states its purpose, who owns the decision, what evidence must exist, what practice is prohibited, what triggers escalation, **what AI may never decide, approve or certify** — and a compliance test an assessor can perform.

They are certification requirements established by the Institute. They are not legislation, and nothing here is legal, tax or accounting advice. External standards are named and described in the Institute's own words, never reproduced.

THE PRINCIPLE BEHIND ALL OF IT

**AI proposes.
The professional
verifies, decides
and remains accountable.**

The full outline and the three Bodies of Knowledge — projectcontrolsinstitute.org